5. EXPERIMENTAL SETUP & MEASUREMENT RESULTS

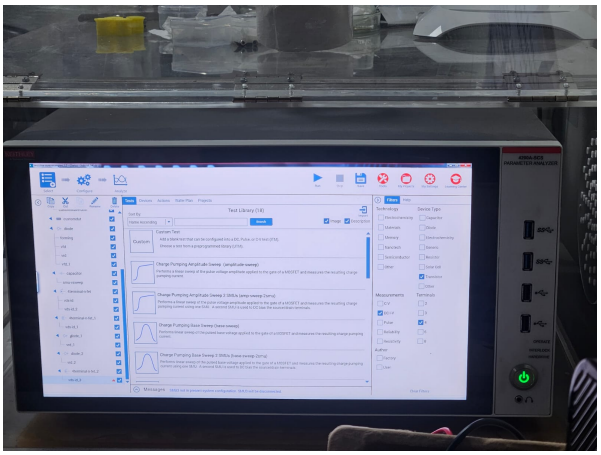


Figure 1: Keithley 4200A-SCS Parameter Analyzer

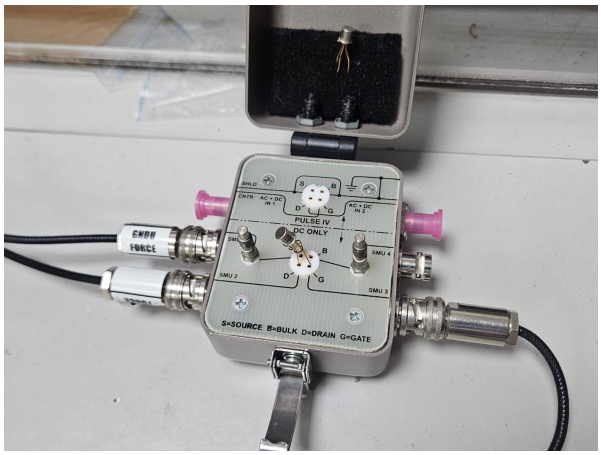


Figure 2: MOSFET Device Under Test (DUT) Fixture

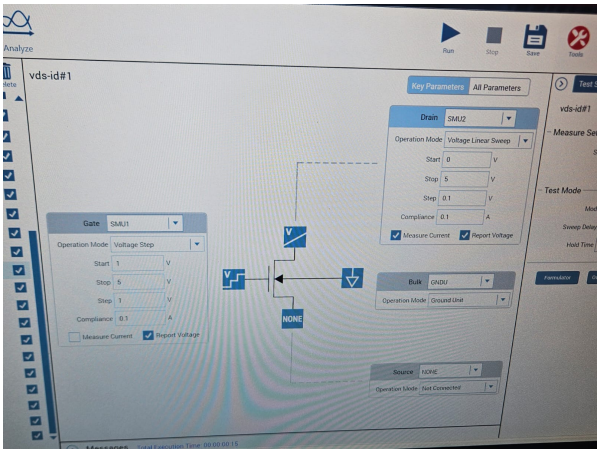


Figure 3: Clarius+ Voltage Sweep Configuration Interface

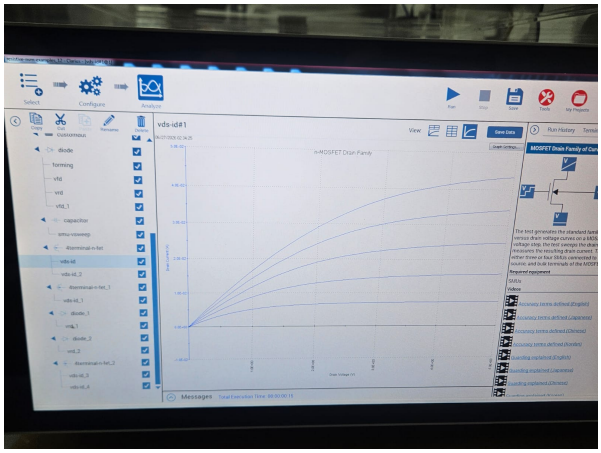


Figure 4: Measured MOSFET Output Characteristics (I_D vs. V_{DS})

5. EXPERIMENTAL SETUP & SPECTRAL ACQUISITION

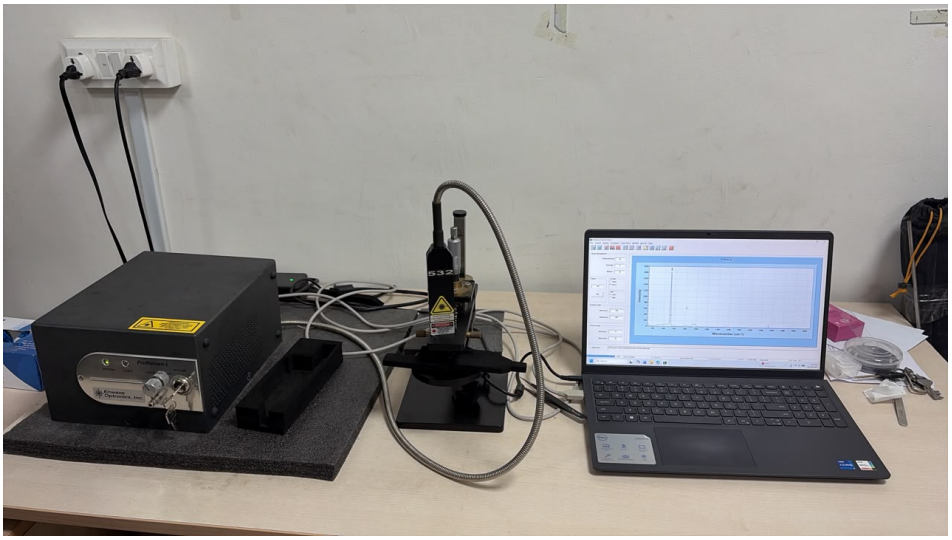


Figure 1: Enwave ProRaman-L Spectrometer System Setup

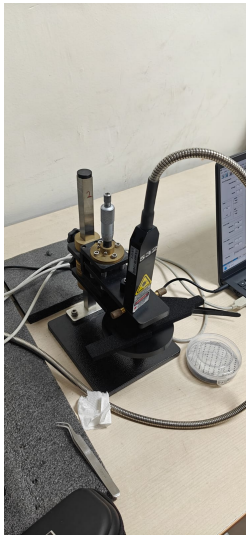


Figure 2: 532 nm Laser Excitation Probe on Silicon Wafer

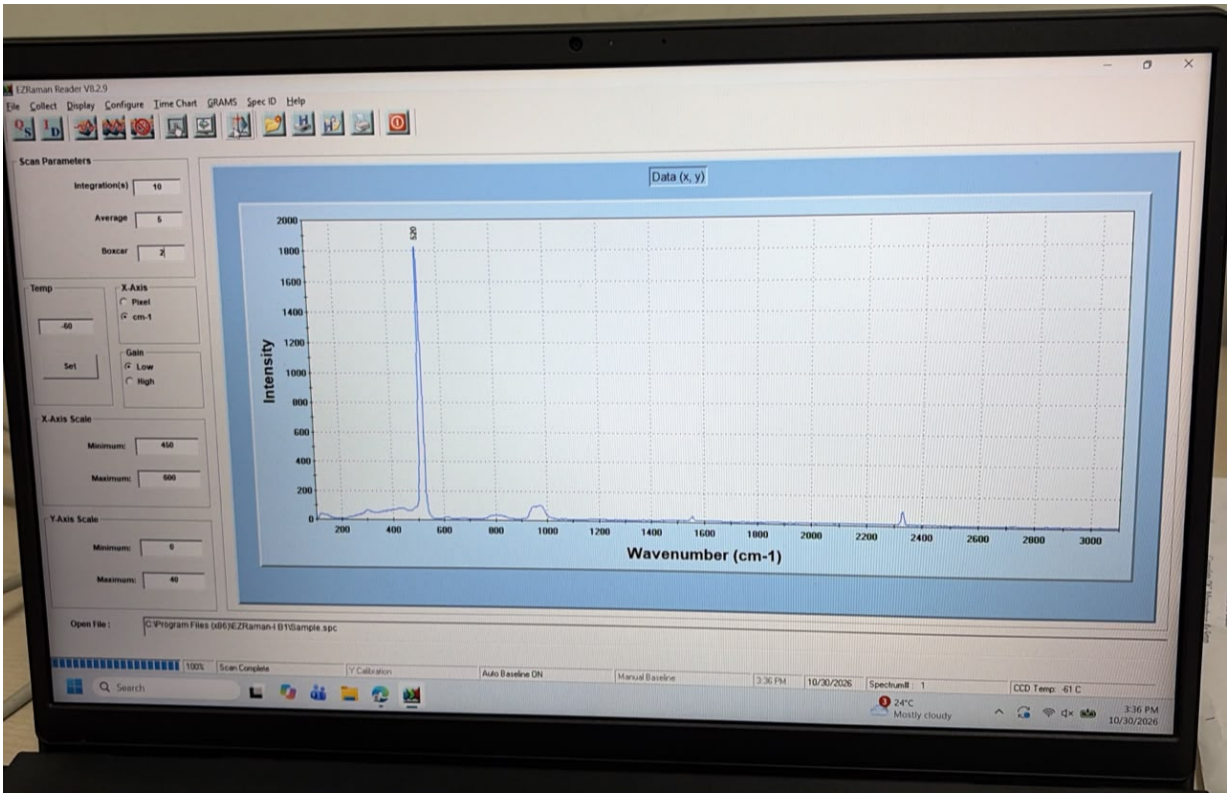


Figure 3: EZRaman Software Displaying Characteristic Silicon Peak at ~520 cm^{-1}

5. REPLACEMENT GATE FABRICATION RESULT

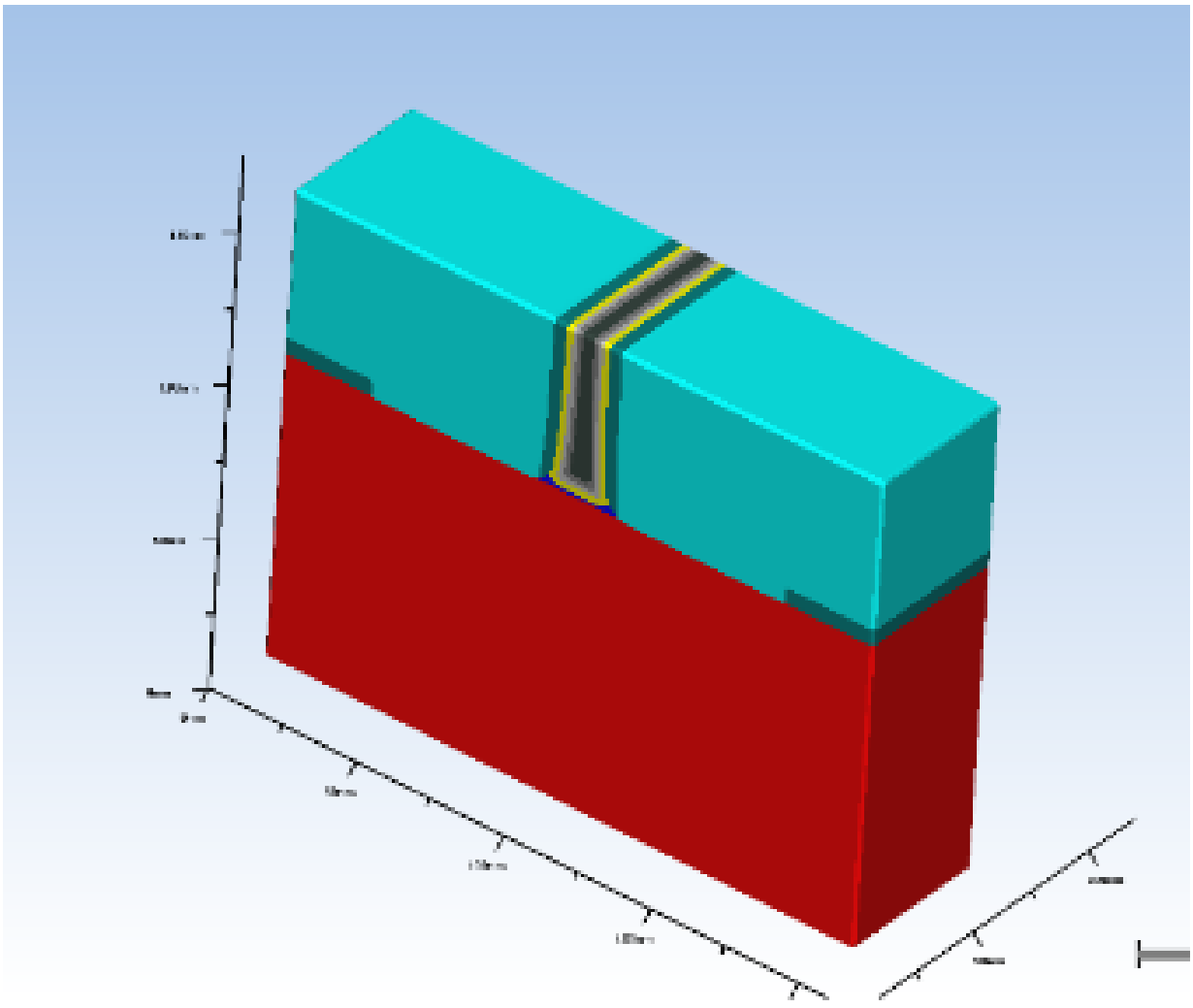


Figure 1: Final 3D Structure of Replacement Metal Gate (RMG) Process

7. ALUMINIUM CONCENTRATION FIELD

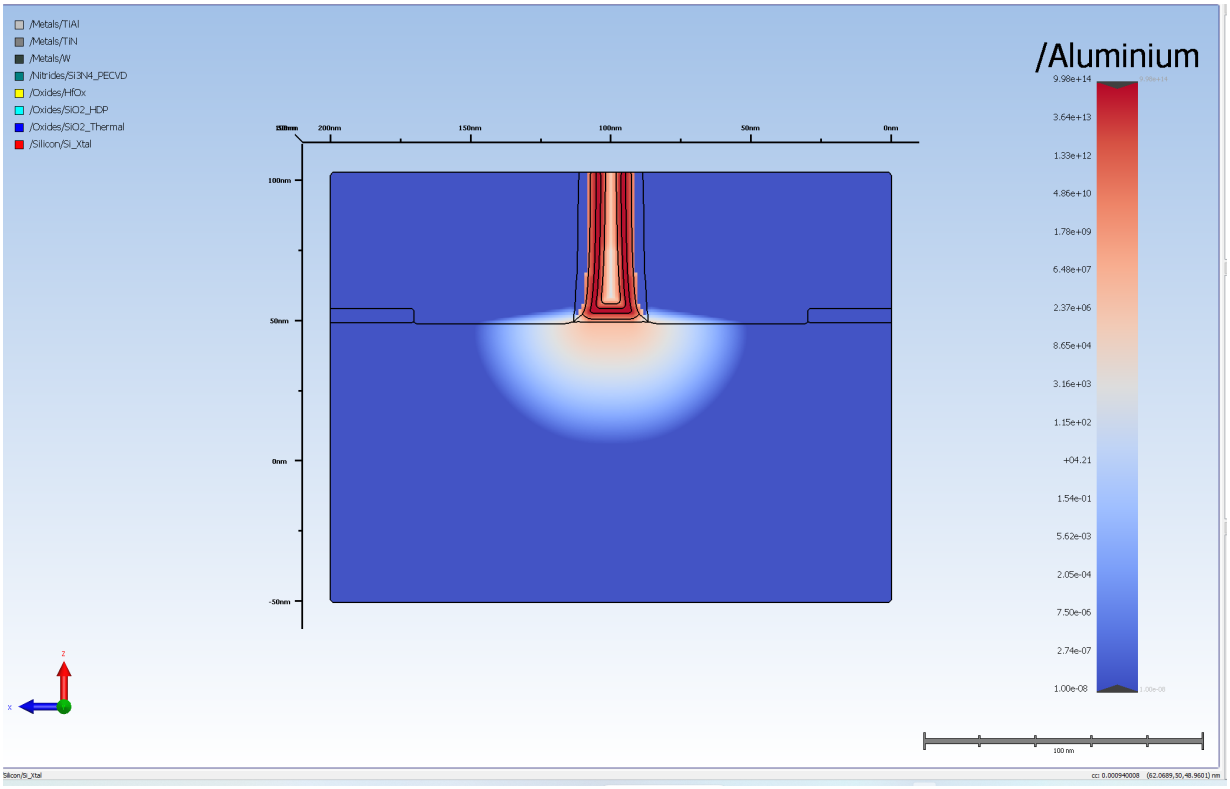


Figure 2: Aluminium Concentration Field Color Map After Dummy Annealing

9. GATE CD VIRTUAL METROLOGY RESPONSE

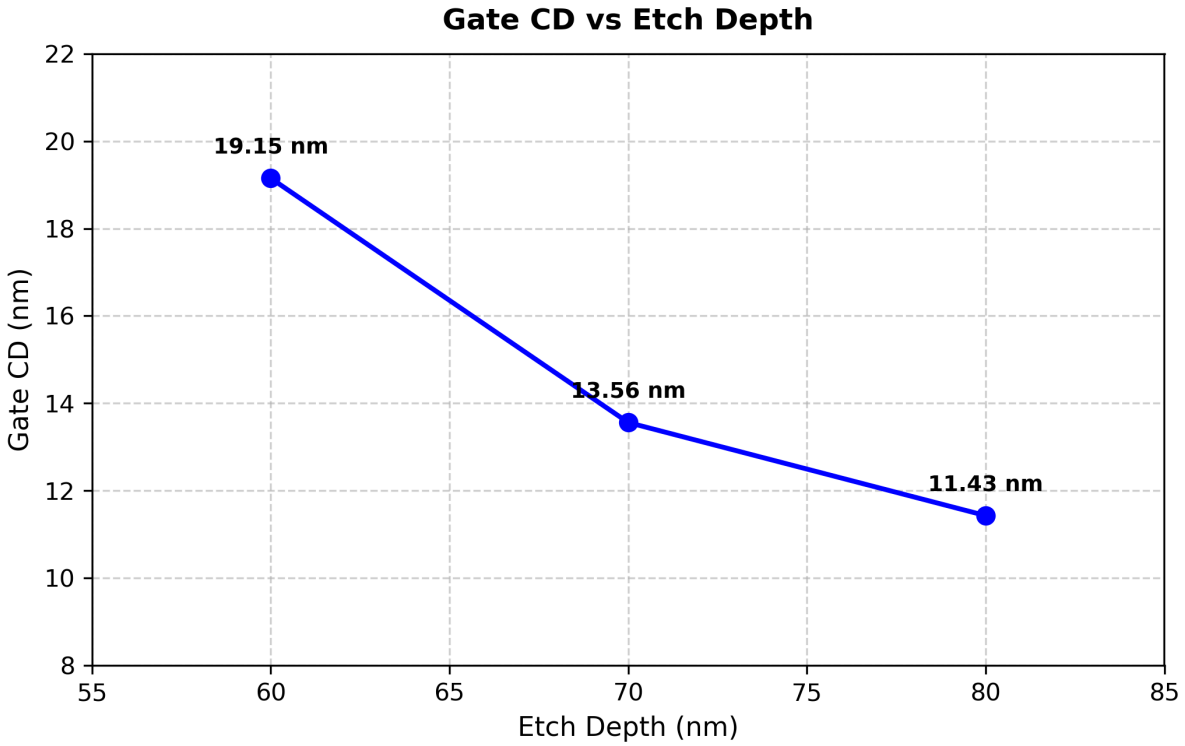


Figure 3: Gate Critical Dimension (CD) vs. Gate Etch Depth

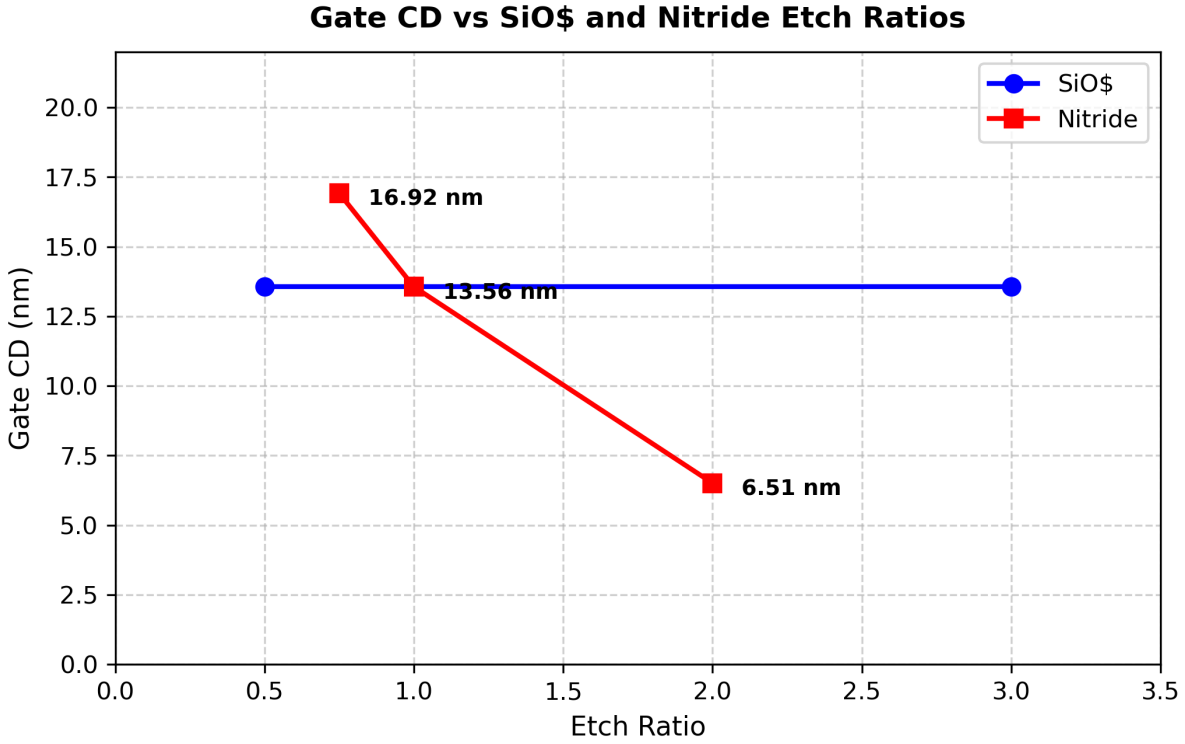


Figure 4: Gate Critical Dimension (CD) vs. SiO2 and Nitride Etch Ratios

11. ALUMINIUM PROFILE: 5 NM LITHO BIAS

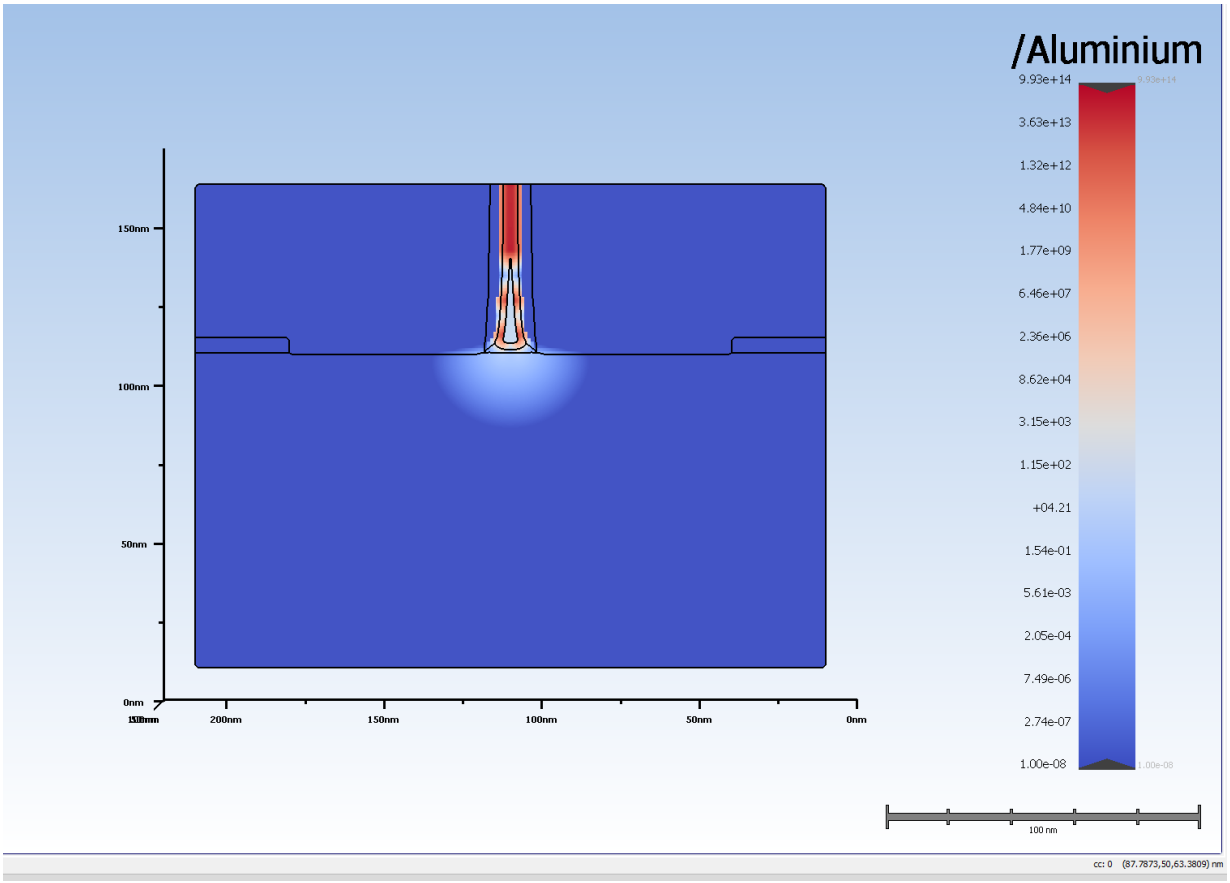


Figure 5: Aluminium Concentration Field After 5 nm Lithography Bias

12. GATE CAVITY VOID DEFECT ANALYSIS

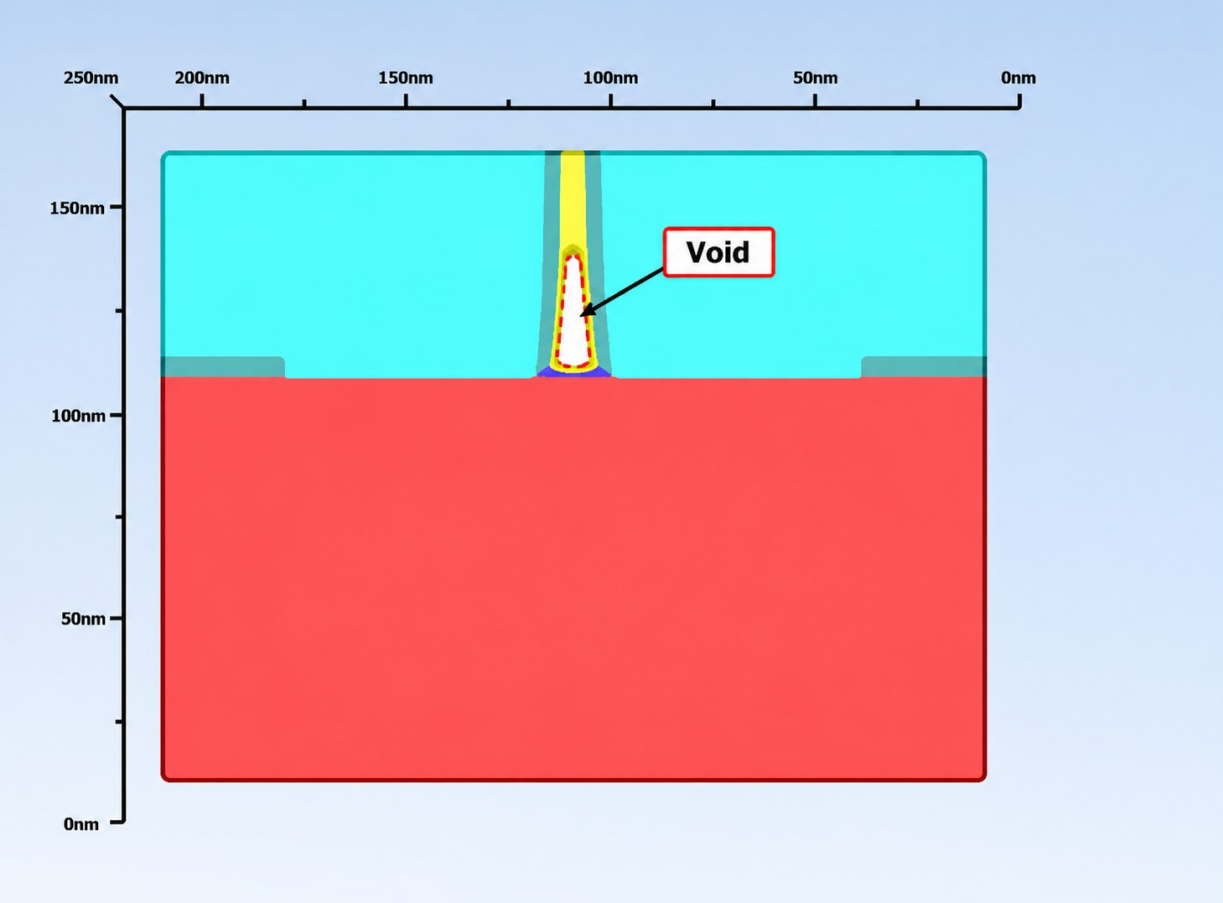


Figure 6: Void Defect Formation Inside Gate Cavity (SiO2 Etch Ratio = 2)

13. ALUMINIUM PROFILE: 10 NM X-OVERLAY

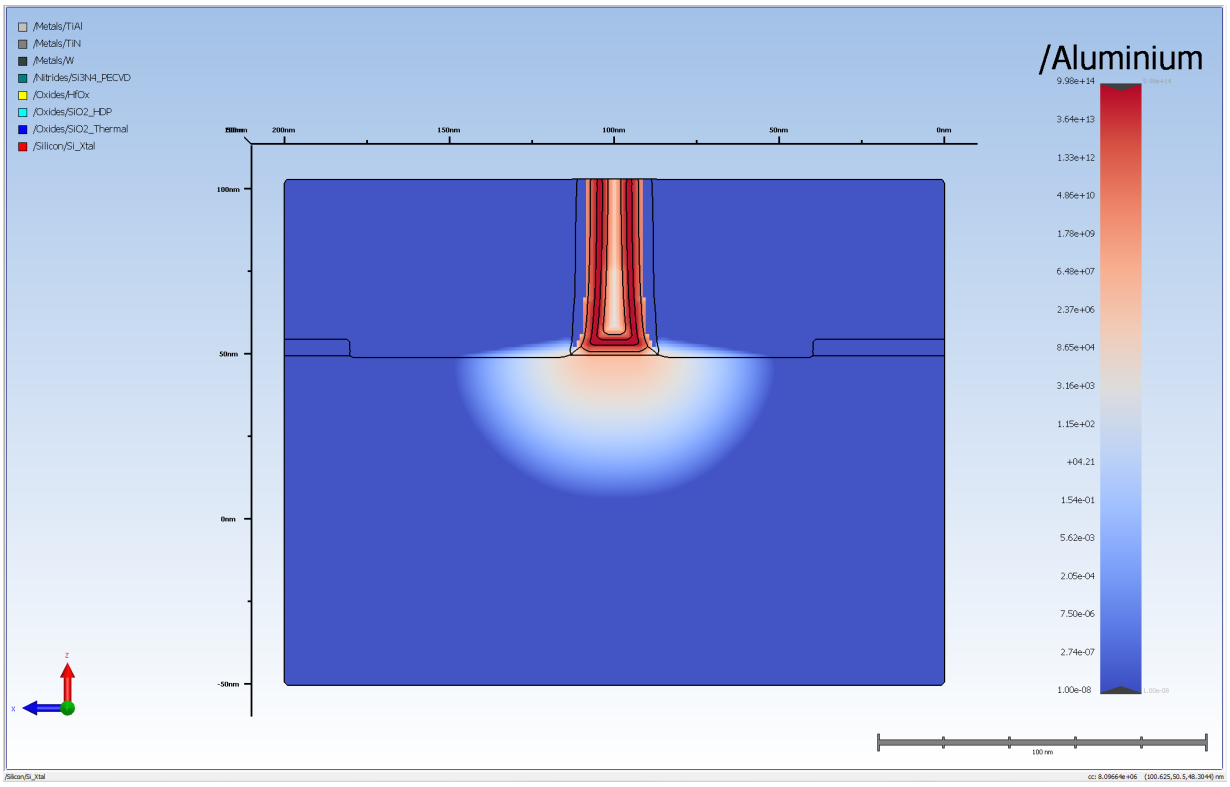


Figure 7: Aluminium Concentration Field After Source_Drain2 Mask & 10 nm X-Overlay